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Copper-Catalyzed Stereoselective Defluoroalkylation or -arylation of *gem*-Difluoroallenes To Form Monofluoroallenes under Mild Conditions

Yaoming Tan[+],[a] Kailun Chen[+],[a] Jiaxin Hu,[a] Sen Lin,[a] and Tao Wu*[a, b]

Herein, we demonstrate the formation of monofluoroallenes through copper-catalyzed defluoroalkylation or defluoroarylation of *gem*-difluoroallenes using affordable and readily available Grignard reagents. Additionally, alkyl lithium or alkyl zinc reagents proved successful in these transformations. Under mild conditions, monofluoroallenes can be converted into fluorinated olefinic compounds or non-fluorinated alkyne compounds. Initial mechanistic studies suggest that the reaction may start with an organocopper reagent and undergo a rapid β-F elimination step. Although the current enantiomeric excess is low, chiral ligands could facilitate the asymmetric transformation of this reaction.

Introduction

The incorporation of a fluorine atom in an organic molecule can lead to substantially altered properties, including biological activity, physical and chemical parameters, as well as material performances, as compared to a non-fluorinated compound.[1] Significant progress has been made in the synthesis of alkyl, alkenyl, aryl, or acyl C–F bonds,[2] but the development of compounds containing allenyl C–F bonds is still in its infancy, perhaps due to concerns about the stability of such species.[3] Due to its promising conversion capability based on allenyl functional groups,[4] monofluoroallene can serve as a single building block for the synthesis of C–F compounds bearing alkyl, alkenyl, alkynyl, carbonyl, carboxylic acids, and other valuable functional groups. Consequently, the development of simple and efficient methods to construct allenyl fluorides is of synthetic importance. Furthermore, as allenyl groups become more prevalent as crucial building blocks in numerous drug molecules, organic ligands, and materials chemistry molecules,[5] this novel fluorinated allenyl moiety provides new opportunities for molecular design and editing in these fields.

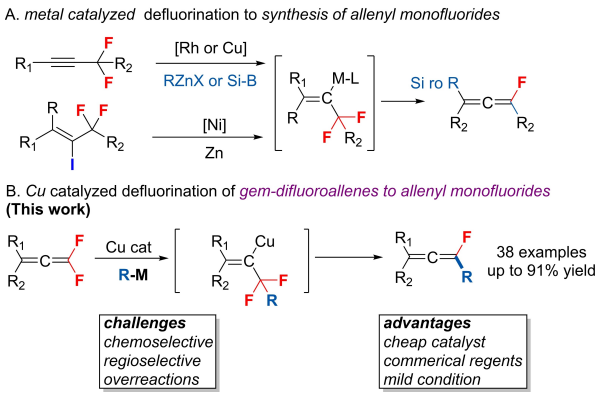
Hayashi[6] and Toste[7] independently reported the metal-catalyzed defluoroalkylation and defluorosilylation of propargyl *gem*-difluorides to monofluoroalkenes (Scheme 1A top). Our group also discovered the nickel-catalyzed defluorination of iodoallyl *gem*-difluorides under reducing conditions to obtain allenyl monofluorides (Scheme 1A bottom).[8] In these reactions, alkenyl metal species serve as central intermediates, whose β-F elimination produces the ultimate monofluorinated product. Hammond and co-workers presented an alternative precursor, *gem*-difluoroallenes, that facilitated the production of monofluoroallenes through their defluorinated nucleophilic substitution reaction. However, this process necessitated a significant excess of copper salts and only two examples have been reported.[9] Inspired by the currently extensively studied catalytic defluorinative functionalization reactions of *gem*-difluoroalkenes,[10] we are attempting to implement a catalytic version of the Hammond's strategy and explore the substrate range and asymmetric trials of the method. We hereby report on the synthesis of monofluoroallenes upon a copper-catalyzed

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Scheme 1. Methods to synthesize allenyl monofluorides.

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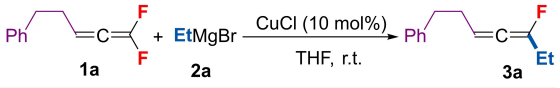
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defluorination of *gem*-difluoroallenes in line with the above proposal (Scheme 1B).[11]

Results and Discussion

We began our hypothesis with *gem*-difluoroallenes **1a** and EtMgBr **2a** under metal catalysts (Table 1). The optimal conditions were finally obtained by systematic examination of the metal catalyst, ligand, base and other reaction parameters such as temperature, solvent, and time, providing the desired monofluoroallene **3a** in 99% GC yield and 88% isolated yield (entry 1). Control experiments showed that copper catalysts played important roles in the reactions. Other metal catalysts, such as palladium or nickel, were ineffective in this transformation (entries 2 and 3). Copper catalysts, other than CuCl, could also catalyze this reaction, but with less efficiently (entries 4 and 5). Notably, **3a** was also obtained using copper(II) salt catalysts (entry 6). Adding a ligand is not advantageous to the reaction and diminishes the yield (entries 7 and 8). The low temperatures slowed down the reactions and gave a decreased yield (entries 9 and 10). Solvents were proven to be one of the most important factors affecting the reactions. Toluene or Hexane, instead of THF as the solvent, would drastically decrease the yield and give **3a** with 10% or 27% GC yield respectively (entries 11 and 12). The amount of Grignard reagent used was another critical factor affecting the efficiency of the conversion. Excessive amounts of EtMgBr probably would cause further conversion of the product of **3a** (entry 13). Lower catalyst loading would consequently lead to lower yield of desire monoallenyl fluoride and a significant amount of **1a** was recovered (entry 14). The removal of copper metal completely shut down the reaction (entry 15).

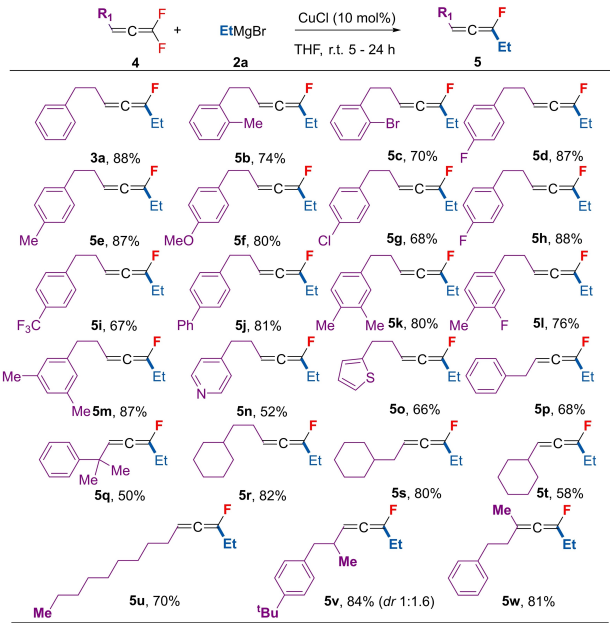
Table 1. Optimization of the reaction.[a]



Entry	Variation from standard conditions	% Yield **3a** [b]
1	none	99%(88%)[c]
2	$PdCl_2$ instead of CuCl	0
3	$NiCl_2$ instead of CuCl	0
4	CuBr instead of CuCl	73%
5	CuTC instead of CuCl	82%
6	$CuCl_2$ instead of CuCl	85%
7	with 20 mol% PPh_3	93%
8	with 10 mol% 1,10-Phen	71%
9	- 30 °C instead of r.t.	86%
10	0 °C instead of r.t.	72%
11	Toluene instead of THF	10%
12	n-Hexane instead of THF	27%
13	1.5 equiv EtMgBr used	48%
14	5 mol% CuCl used	27%
15	no CuCl	n.d.

[a] Reaction condition: **1a** (0.1 mmol), EtMgBr **2a** (1.1 equiv), THF (1 mL). [b] Percentages represent GC yields of **3a** using n-tetradecane as internal standard. n.d. = not detected. [c] Value in parentheses is isolated yield of **3a**.

With an optimal process in hand, we next investigated the scope of the gem-difluoroallenes component in this transformation (Scheme 2). Various trisubstituted monofluoroallenes were smoothly obtained with satisfactory yields by the combination of corresponding allenes **4** and EtMgBr in the presence of the copper catalyst. A wide array of substituted phenylethyl difluoroallenes, bearing methyl at the *ortho*-, *meta*-, and *para*-positions, or electron donating group-OMe, or electron withdrawing group-CF_3, or halide group at the *para*-positions, worked well, affording expected defluoroethylation allenes **5b–5i** in good to excellent yields. Noteworthy, bromo- and chloro-substrates **4c** and **4g**, which were easily coupled with Grignard regents under transition metal catalysts, were tolerated and retained under our reaction conditions, transferring to ethylated allenes **5c** and **5g** in 70% and 68% yields respectively. Phenyl and dimethyl substituted phenylethyl allenes were also applied to reaction conditions and to furnish the desired products **5j**, **5k** and **5m** in moderate to excellent yields. Pyridine or thiophene tethered allenes **4n** and **4o** engaged in the reactions, and successfully defluorinated and alkylation to furnish the corresponding allenyl fluorides **5n** and **5o** in moderate yields. Benzylated *gem*-difluoroallenes, also worked well and afford **5p** and **5q** in contented yields. 1-cyclohexylethylated, 1-cyclohexylmethylated, cyclohexylated allenes, a series of substrates accompanied by decreasing carbon chains, were tested, and gently transferred to anticipated products **5r–5t** in good yields. Prolonged carbon chain monofluoroallene **5u** was also achieved at 70% yield. Racemic substrate **4v** with a tertiary carbon, afforded two unseparatable diastereoisomers at 84% yield with 1.6:1 dr, which was attributed to generation of the axial chirality of allenes via defluoroalkylation process. It is gratifying to note that tetrasub-



Scheme 2. Scope of *gem*-difluoroallenes. Reaction conditions: CuCl (0.02 mmol, 10 mol%), **4** (0.2 mmol, 1 equiv), **2a** (0.22 mmol, 1.1 equiv) in THF (2 mL), stirred for 5–24 hours at room temperature, average isolated yield after two runs.

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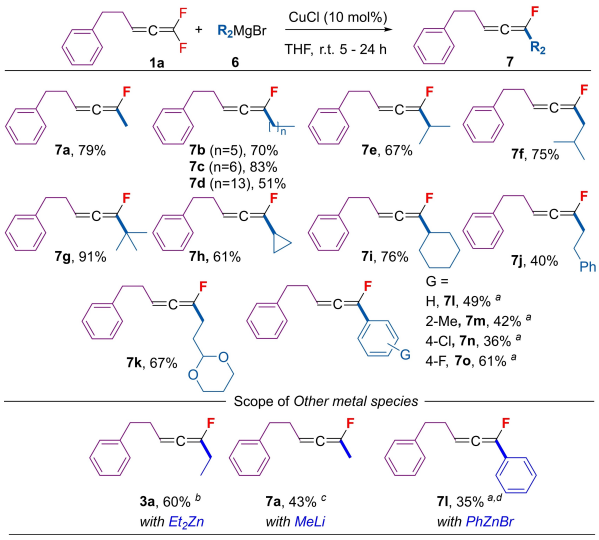
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stituted *gem*-difluoroallene **4w** could also be applied to our reaction conditions and defluoroalkylation product **5w**, persubstituted monofluoroallene was acquired with 81% yield.

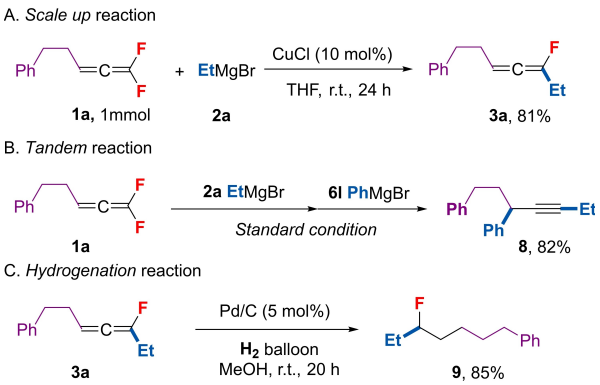
We next investigated different Grignard reagents displaying a variety of electronic and steric properties (Scheme 3). All types of alkyl Grignard reagents could be used to afford the corresponding defluoroalkylation monofluoroallenes in moderate or excellent yields. Long carbon chain Grignard species, tetradecylmagnesium chloride, smoothly reacted with *gem*-difluoroallenes **1a**, and tetradecylated desired product **7d** was afforded at 51% yield. Other commercial alkyl magnesium compounds, such as *iso*-propyl, *iso*-butyl, and *tert*-butyl, also worked well in these transformations, leading to allenes **7e**–**7g** respectively. A cyclopropyl group also could be introduced into monofluoroallene **7h** with related metal species, in which ring-opening products were not detected. Acetal-protected carbonyl containing Grignard compound, **6k** engaged in the reaction and was transferred to allenyl fluoride **7k** in 67% yield, easy to remove the protecting group and return to the aldehyde group for more relevant transformations. Aryl magnesium bromides or chlorides, were also examined in this process. Moderated yields of defluoroarylated allenes **7l**–**7o** were achieved under low temperature, stemmed from the low stability of these aryl-allenyl fluorides. Other metal species, also were considered and applied into this system. Organozinc reagents, and organolithium compounds, were participated in the reaction instead of Grignard reagents and the corresponding products were obtained, nevertheless with less efficiency.

Because of the rich reactivity of the allene group, we expect that these monofluoroallenes products could be converted into more useful synthons (Scheme 4). First, it is simple to expand our reactions to 1 mmol scale, just with a little lower yield of product **3a**. Second, alkynes compounds can be obtained simply by tandem reactions from **1a**. Alkyne **8** was afforded by addition of PhMgBr into the mixture of our reaction without the separation of monofluoroallene **3a**.[12] It is a new way to synthesis kinds of alkynes by simple operation of changing substitutes on *gem*-difluoroallenes and Grignard reagents. Allenyl fluoride **3a**, a new sp^2 C–F bond compound, was transferred to alkyl fluoride **9** by palladium-catalyzed hydrogenation at atmospheric pressure and room temperature with 85% yield,[13] in which C–F bond was retained.

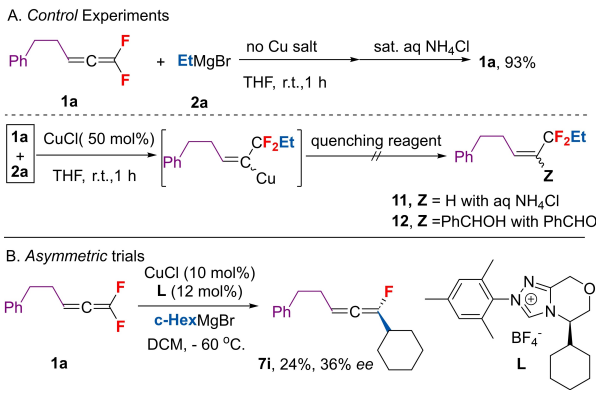
We designed several controlled experiments to try to explore the mechanism of the reaction (Scheme 5). **1a** was almost completely recovered in the absence of copper catalyst.[14] After mixing the substrates **1a** and **2a** with 0.5 equivalent copper salt for one hour, the reaction was quenched by adding saturated aqueous ammonium chloride solution or benzaldehyde respectively.[15] Expected allyl *gem*-difluoride **11** and allyl alcohol **12** were not detected when analyzing of the residues of the above quenched reactions, other than 72% yield of **3a** was obtained.[16] These results seem to imply that the reaction proceeds through the alkyl copper species and gives the final product by its speedy β-F elimination via an alkenyl copper intermediate.[17] We also tried asymmetric reactions, although moderate enantioselectivity could be obtained, but the conversion of this reaction was low.[18]



Scheme 3. Scope of Grignard reagents and other metal species. Reaction condition: CuCl (0.02 mmol, 10 mol%), **4** (0.2 mmol, 1 equiv), **6** (0.22 mmol, 1.1 equiv) in THF (2 mL), stirred for 5–24 hours at room temperature, average isolated yield after two runs. [a] the reactions were run at −60 °C. [b] 1.1 equiv. Et_2Zn was used instead of EtMgBr. [c] 1.1 equiv. MeLi was used instead of MeMgBr. [d] 1.1 equiv. PhZnBr was used instead of PhMgBr.



Scheme 4. Scale-up and derivatizations.



Scheme 5. Some mechanistic considerations.

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Conclusions

We have developed a copper-catalyzed synthesis of monofluoroallenes via defluorination of *gem*-difluoroallenes under mild conditions. This procedure generated various alkylated or arylated allenyl monofluorides with moderate to excellent yield using commercial Grignard reagents. Other metal species, including organozinc and organolithium compounds, can also be used with these new catalytic approaches. Monoallenyl fluorides are crucial precursors for organic synthesis as they can be easily converted to alkynes or alkyl fluorides by a simple process. Mechanistic studies have indicated that the reaction may involve organocopper species and alkenyl copper intermediates, resulting in the desired products through β-F elimination. Although attempts at inducing asymmetric reactions with chiral ligands were unsuccessful, they are currently the focus of our ongoing research.

Experimental Section

To an oven-dried tube equipped with a magnetic stir bar, CuCl (2 mg, 0.02 mmol) and THF (0.8 mL) were added and cooled to −60 °C under nitrogen. Subsequently, Grignard reagent **2a** or **6** (1.1 equiv, 0.22 mmol) was added and stirred for 5 minutes at this temperature. Then, a THF solution of allene **1a** or **4** (1 mL, 0.2 mmol/1 mL, 1 equiv) was added. The cold bath was removed, and the reaction was stirred at room temperature for 5 hours. The reaction was quenched with methanol and saturated NH_4Cl aqueous solution at 0 °C, and purified by silica gel column chromatography to afford the desired products.

Supporting Information

Additional references cited within the Supporting Information.[3e,19]

Acknowledgements

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: copper catalysis · β-F elimination · *gem*-difluoroallenes · Grignard reagents · monofluoroallenes

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